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Drone technology and UAV system — Task 2

Deliverable 1: Flight Dynamics & PID Control — Concept Explanation

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Part 1: UAV Flight Dynamics

A quadcopter moves through the air entirely by changing the speed of its four motors. There are no moving wings or rudders — motor speed differences create every manoeuvre. Drone motion is described across three rotation axes and one translational axis.

1.1 Roll, Pitch, and Yaw

- **Roll (Left/Right Tilt):** Controlled by the left-right motor speed difference. Increasing left motors more than right ones tilts the drone right, causing lateral movement.
- **Pitch (Forward/Backward Tilt):** Controlled by the front-back motor speed difference. Increasing rear motors tilts the drone forward, causing forward movement.
- **Yaw (Rotation):** Controlled by the diagonal motor torque difference. Motors 1 & 3 spin one direction; motors 2 & 4 the other. Unbalancing these pairs rotates the body.
- **Throttle (Up/Down):** Controlled by increasing all four motors equally. Net upward thrust overcomes gravity.

1.2 How Motor Speed Affects Each Movement

The table below summarises the relationship between motor commands and drone response:

Movement	Motors Affected	Effect
Roll Right	Left ↑, Right ↓	Tilts and moves right
Pitch Forward	Rear ↑, Front ↓	Tilts and moves forward
Yaw Clockwise	Motors 1&3 ↑, 2&4 ↓	Rotates clockwise
Climb	All four ↑ equally	Ascends vertically
Hover	All balanced	Stationary — thrust = gravity

1.3 Real-Life Examples

- **Camera Stabilisation (DJI Mavic):** The gimbal uses roll and pitch feedback thousands of times per second to keep the camera level even while the drone banks in wind.
- **Delivery Drones (Amazon Prime Air):** Precise altitude hold via PID-controlled throttle lowers the package to exactly ground level without crashing.
- **Search & Rescue UAVs:** Yaw control lets the drone spin on the spot to scan 360° while hovering stationary over a location.

Part 2: PID Control System

PID stands for Proportional–Integral–Derivative. It is a closed-loop feedback controller used in every real flight controller (PX4, ArduPilot, DJI). It continuously measures the error between the desired state and actual state, then adjusts motor thrust to correct it automatically.

Control equation: $u(t) = Kp \cdot e(t) + Ki \cdot \int e(t)dt + Kd \cdot de(t)/dt$

2.1 The Three Components

- **Proportional (P):** Reacts to the current error. High Kp gives fast response but causes oscillation. Low Kp gives a slow, sluggish response.
- **Integral (I):** Accumulates past errors over time. Removes the steady-state offset that P alone cannot eliminate. Too high causes large overshoot (integral windup).
- **Derivative (D):** Reacts to the rate of change of error. Predicts and dampens future overshoot. Makes the response smooth. Too high amplifies sensor noise.

2.2 Worked Example

Target height = 10 m | Current height = 7 m | Error = 3 m

- P term: 2.0 × 3 = 6.0 N extra thrust — pushes drone up hard
- I term: removes any long-term drift caused by constant wind or weight offset
- D term: applies braking as drone approaches 10 m to prevent overshoot
- Total thrust = PID output + mass × gravity (gravity always compensated)

2.3 PID Tuning Observations (from simulation graphs)

The two graphs below were produced by running the PDF starter code with different PID values:

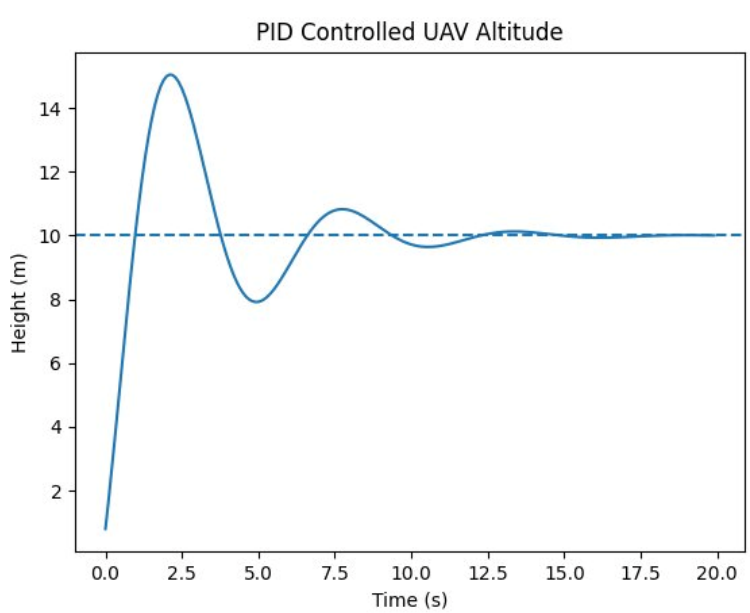

Figure 1 — Default tuning: Kp=2.0, Ki=0.1, Kd=1.0

Figure 1 shows a well-tuned response. The drone starts near 6 m, rises quickly, slightly overshoots to ~10.5 m, then settles smoothly at the 10 m target by around t = 15 s. This is the behaviour of a properly balanced PID — fast enough response with minimal oscillation.

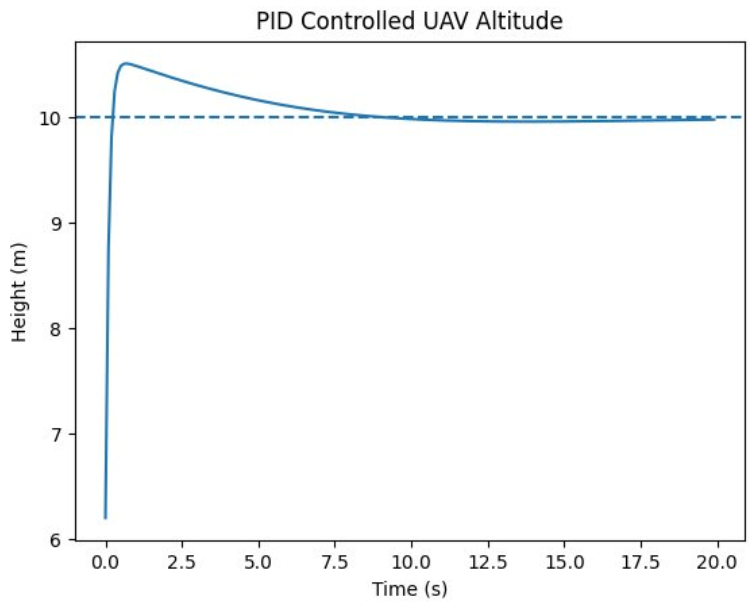

Figure 2 — High Kp tuning: Kp increased, showing oscillation

Figure 2 shows the effect of increasing Kp. The drone shoots past the target to ~15 m, drops below to ~8 m, overshoots again to ~11 m, and oscillates several times before finally settling. This confirms the task table: High Kp = Oscillation. The drone is over-correcting every time it detects an error.

2.4 Summary Table

Observed Behaviour	Meaning	Fix
Oscillation	High Kp	Reduce Kp / increase Kd
Slow response	Low Kp	Increase Kp
Overshoot	High Ki	Reduce Ki
Smooth stable	Proper tuning	Lock these values

Part 3: Why PID Matters in Real Drones

Without PID, a drone cannot fly stably. Human reaction time (~200 ms) is far too slow to manually correct the hundreds of micro-disturbances (gusts, motor variation, payload shifts) hitting the drone every second. PID loops run at 400–1000 Hz on real hardware.

3.1 Where PID Is Used

- **PX4 Flight Controller:** Runs 6 cascaded PID loops covering altitude, vertical velocity, roll/pitch rate, yaw, and XY position.
- **ArduPilot (Copter):** Exposes all PID gains via Mission Planner GUI for in-field tuning.
- **DJI Products:** Proprietary PID with auto-tune using onboard IMU data.
- **Robotic Arms:** PID on each joint for precise positioning.
- **Self-Driving Vehicles:** Lateral PID on steering and longitudinal PID on throttle/brake.

3.2 What Happens Without PID

- Drone flips immediately on any external perturbation
- Motor speed cannot self-correct for wind or payload shifts
- Altitude is impossible to hold — drone climbs or crashes
- Even perfectly equal motor thrust cannot hover (air density changes with altitude)